Adaptive Lookahead Routing, Fault-Tolerant Live Detours, and Belief-State Filtering in Energy-Harvesting Wireless Sensor Networks

Ananthakrishnan R, Santhosh G, and Thilang TV

Department of Electrical and Computer Science Engineering, School of Electrical Engineering (SELECT)
Vellore Institute of Technology (VIT), Vellore 632014, Tamil Nadu, India
Undergraduate B.Tech Researchers — Reg. Nos: Ananthakrishnan R (25BEL0005), Santhosh G (25BEL0016), Thilang TV (25BEL0036)
Public Software Repository: https://github.com/santhoshvellore7119-web/WSN_proj (Branch: master)

Abstract—Wireless Sensor Networks (WSNs) powered by ambient renewable energy harvesting (micro-photovoltaics, RF rectennas, thermal gradients) challenge the foundational assumption of classical network optimization: that stored electrochemical energy monotonically drains over time ($E_i(t_1) \ge E_i(t_2)$ for $t_1 \le t_2$). Standard routing protocols—including Dijkstra's shortest path, standard LEACH clustering, and static bottleneck Bellman-Ford—evaluate candidate relay nodes using static energy snapshots captured at the start of a transmission round ($t=0$). In renewable networks, this static perspective causes severe structural pathologies: *false rejection* of recharging relays (nodes whose initially depleted reserves replenish during preceding multi-hop packet transit delay) and *accelerated burnout* of occluded, shaded nodes. Furthermore, when continuous battery telemetry is throttled to conserve transceiver energy, hidden mid-round relay brownouts trigger in-flight packet drops.

In this paper, we develop a lightweight, deterministic algorithmic framework tailored for resource-constrained microcontrollers: (1) **3D Time-Augmented Dynamic Programming ($dp[v][h][t]$):** A polynomial-time routing algorithm over a time-expanded DAG that plans multi-hop paths around projected solar replenishment, guaranteeing a provable 2ϵ-approximation bound under bounded stochastic harvest variance $|\xi| \le \epsilon$; (2) **Disjoint-Set Union (DSU) Live Detour Recovery:** An in-flight fault-repair mechanism utilizing Union-Find with path compression and rank optimization that splices detour bridges around brownout relays in $O(\text{deg}(u)\alpha(V))$ time, delivering a $3.9\times - 6.1\times$ execution speedup over full DP recomputation with 0% packet loss up to a 30% node failure rate; (3) **Probabilistic Belief Routing Under Telemetry Constraints:** Truth/belief decoupling via a 24-bin discrete Histogram Bayesian Filter with exact mass boundary clipping at 0 and C_i (eliminating Gaussian tail leakage), Sheppard's second-order variance correction, and log-domain Max-Product Belief DP with recharge waiting deadlines ($w \in [0, D]$); (4) **Physical Realism Enhancements:** Comprehensive characterization of first-order electronic reception overhead ($E_{\text{rx}} = k \cdot E_{\text{elec}}$), log-normal shadow channel fading ($\sigma_{\text{shadow}} \in [0, 5]\text{ dB}$), continuous Sigmoid Packet Reception Rate (PRR), MAC layer retransmissions, and ADC sensor quantization jitter.

The proposed framework is verified through an automated 75-test unit suite (100% pass rate in 1.51s) and extensive multi-seed statistical benchmarks ($N=30$ seeds, 350 rounds). Under scarce solar irradiance, our belief-driven routing closes up to 81.5% of the omniscient oracle delivery gap while reducing Expected Calibration Error from 0.039 to 0.019 over stale telemetry. In a real-world case study on the Great Duck Island habitat topology ($N=32$ motes, 60% spruce canopy shade), our framework sustains 100% mote survival with zero node deaths across 500 rounds (+15.5% mission extension over LEACH).

Index Terms—Energy Harvesting, Wireless Sensor Networks, Dynamic Programming, Disjoint-Set Union, Fault-Tolerant Routing, Bayesian Belief Filtering, Calibration Error, Telemetry Optimization, Radio Dissipation Physics.

I. INTRODUCTION

Wireless Sensor Networks (WSNs) are deployed across environmental monitoring, precision agriculture, structural safety, and industrial automation [1], [2]. Historically, sensor motes have been constrained by primary electrochemical batteries (e.g., Lithium coin cells). In battery-powered networks, stored energy decreases monotonically with every sensing, processing, and transmission cycle: $E_i(t_1) \ge E_i(t_2) \; \forall\; t_1 \le t_2$. Under this monotonic invariant, classical routing protocols—such as Dijkstra's algorithm, A* search, and standard LEACH clustering [3]—evaluate candidate forwarders based on static battery snapshots at round inception ($t = 0$). If a candidate relay has low battery at $t = 0$, these protocols permanently prune it from the forwarding tree to avoid dropping packets.

Recent advances in micro-photovoltaic cells, piezoelectric harvesters, and supercapacitor energy storage have enabled modern sensor nodes to operate indefinitely on ambient renewable energy [4], [5]. This introduces a fundamental paradigm shift: **battery energy is non-monotonic**. A sensor node whose reserves are nearly exhausted at the beginning of a cycle can harvest substantial solar replenishment while preceding data packets travel through earlier multi-hop hops.

A. Pathologies in Snapshot Protocols

When classical static routing protocols are deployed in renewable energy-harvesting sensor networks, their blindness to incoming energy induces two primary failure modes:

1) False Rejection Pathology: At round inception ($t = 0$), relay u has residual battery $E_u(0) < E_{\text{tx}}$. Classical snapshot algorithms immediately prune node u. However, because packet forwarding across preceding hops incurs non-negligible transmission and queuing delay ($t_i = i \cdot \delta$), node u receives ambient solar irradiance during transit. Discarding node u forces the network onto circuitous detours that burn excess global energy.

2) Shaded Burnout Pathology: Nodes situated beneath dense canopy receive zero solar recharge ($H_v(t) \approx 0$), yet may begin with substantial initial reserves. Snapshot protocols repeatedly select these shaded nodes as forwarders due to their high instantaneous battery levels, driving them into early exhaustion.

3) In-Flight Relay Brownout: Sudden cloud occlusion or voltage droop causes in-flight packet drops. Global recalculation ($O(|E|HT)$) introduces prohibitive latency.

B. Undergraduate Research Contributions

As undergraduate researchers at Vellore Institute of Technology (SELECT), we developed a deterministic, polynomial-time framework grounded in fundamental Data Structures and Algorithms:

1) **3D Time-Augmented DP ($dp[v][h][t]$):** Polynomial-time routing over a time-expanded DAG that plans routes across node, hop, and time dimensions with a proven 2ϵ-approximation bound under bounded arrival variance $|\xi| \le \epsilon$.
2) **DSU Live Detour Recovery:** Local bridge splicing around brownout relays in $O(\text{deg}(u)\alpha(V))$ time, providing a $3.9\times - 6.1\times$ latency reduction over global DP recomputation with zero packet loss.
3) **Probabilistic Belief Routing:** Truth/belief decoupling via an EnergyView protocol and a 24-bin discrete Histogram Belief Filter with exact boundary clipping at 0 and C_i.
4) **Physical Realism:** Characterizes first-order receiver dissipation penalty ($E_{\text{rx}} = k \cdot E_{\text{elec}}$), log-normal shadow channel fading, and initial battery heterogeneity.
5) **Full Reproducibility:** Automated 75-test unit verification suite (100% pass rate in 1.51s) and multi-seed statistical benchmarks ($N=30$ seeds).

II. TRANSCEIVER PHYSICS & CHANNEL PROPAGATION MODEL

A. First-Order Radio Dissipation Physics

Energy dissipation per transmission and reception adheres to Heinzelman's radio model [3]:

$$E_{tx}(k, d) = k\,E_{elec} + k\,\varepsilon_{fs}\,d^{2}\ (d < d_{0})\ \mid\ k\,E_{elec} + k\,\varepsilon_{mp}\,d^{4}\ (d \ge d_{0})$$

$$E_{rx}(k) = k\,E_{elec}$$

where $d_0 = \sqrt{\epsilon_{\text{fs}} / \epsilon_{\text{mp}}} \approx 87.70\text{ m}$. Parameters: $E_{\text{elec}} = 50\text{ nJ/bit}$, $\epsilon_{\text{fs}} = 10\text{ pJ/bit/m}^2$, $\epsilon_{\text{mp}} = 0.0013\text{ pJ/bit/m}^4$, and packet size $k = 4000\text{ bits}$ (500 bytes).

B. The Electronic Reception Overhead (E_{rx}) Trade-Off

In multi-hop routing, each forwarder expends $E_{\text{rx}} = k \cdot E_{\text{elec}} = 0.20\text{ mJ}$. Over distance $d = 60\text{ m}$ ($d < d_0$): Single-hop direct consumes $E_{\text{direct}} = 0.344\text{ mJ}$, whereas two equal 30m hops consume $E_{\text{tx1}} + E_{\text{tx2}} + E_{\text{rx}} = 0.672\text{ mJ}$. Multi-hop relaying is beneficial only when distance exponents (d^4 vs d^2) or ambient solar replenishment overcomes the fixed electronic reception penalty E_{rx}.

C. Log-Normal Shadow Fading & Retransmissions

Under realistic channel fading, Packet Reception Rate (PRR) transitions continuously: $\text{PRR}(d) = [1 + \exp(-(R_{\text{eff}} - d)/(0.15 R_{\text{eff}}) - X_\sigma/3.0)]^{-1}$, where $X_\sigma \sim \mathcal{N}(0, \sigma_{\text{shadow}}^2)$ captures log-normal shadowing in dB. When transmission fails, motes retransmit up to N_{retry} attempts, burning $E_{\text{tx}} \times (1 + \text{retries})$.

TABLE I: Simulation Parameters and Transceiver Calibration

Parameter Description	Symbol	Physical Value & Calibration
Network Terrain Dimensions	`W × H`	100 m × 100 m (2D plane)
Sensor Deployment Density	`N`	50 stationary motes
Base Station Coordinates	`v_BS`	(50.0 m, 50.0 m) [Center sink]
Maximum Radio Range	`R_tx`	48.0 meters (or 35.0 m)
Transceiver Electronics Dissipation	`E_elec`	50.0 nJ/bit (TI CC2420)
Free-Space RF Amplifier Constant	`ε_fs`	10.0 pJ/bit/m^2 (d < d0)
Multipath RF Amplifier Constant	`ε_mp`	0.0013 pJ/bit/m^4 (d ≥ d0)
Crossover Distance Threshold	`d_0`	87.70 meters
Data Packet Payload	`k`	4000 bits (500 bytes)
Mote Battery Capacity	`C_max`	0.50 Joules
Receiver Circuit Overhead per Hop	`E_rx`	0.20 mJ (= k · E_elec)

III. PROPOSED ALGORITHMIC FRAMEWORK

A. 3D Time-Augmented Dynamic Programming ($dp[v][h][t]$)

We define the 3D table $dp[v][h][t]$ representing the maximum bottleneck residual energy along a path reaching node v after h hops at time offset t:

$$dp[v][h][t] = \max_{u \in nbr(v)} \min(\ dp[u][h-1][t-\delta],\ \ E_{proj}(v, t_{curr} + t) - E_{rx}\ )$$

subject to $dp[u][h-1][t-\delta] \ge E_{\text{tx}}(k, d(u, v))$, with projected battery bounded by capacity C_v: $E_{\text{proj}}(v, t_{\text{curr}} + t) = \min(C_v, E_v(t_{\text{curr}}) + \int H_v(\tau) d\tau)$. Base initialization: $dp[s][0][0] = E_s(t_{\text{curr}})$, and $dp[v][h][t] = -\infty$ elsewhere.

> **Theorem 1 (Approximation Bound Under Bounded Variance):** Let P^* be the optimal path selected by an omniscient offline oracle. Under bounded stochastic harvest error $|\xi_v(t)| \le \epsilon$, the path $P_{\text{Time-DP}}$ computed by 3D Time-Augmented DP satisfies: $\text{Bottleneck}(P_{\text{Time-DP}}) \ge \text{Bottleneck}(P^*) - 2\epsilon$.

Proof: Let $\hat{B}(P)$ denote projected bottleneck energy and $B(P)$ realized physical energy. Since $|\xi_v(t)| \le \epsilon$, for all paths P: $|B(P) - \hat{B}(P)| \le \epsilon$. Because Time-Augmented DP guarantees exact optimality over projected values: $\hat{B}(P_{\text{Time-DP}}) \ge \hat{B}(P^*)$. Applying the triangle inequality: $B(P_{\text{Time-DP}}) \ge \hat{B}(P_{\text{Time-DP}}) - \epsilon \ge \hat{B}(P^*) - \epsilon \ge (B(P^*) - \epsilon) -

\epsilon = B(P^*) - 2\epsilon$. ■

B. Disjoint-Set Union (DSU) Live Detour Recovery

When an intermediate relay node depletes unexpectedly mid-round: (1) Upstream predecessor $u = v_{\text{fail}-1}$ detects that next-hop relay v is inoperable; (2) Predecessor u queries its 1-hop physical neighbors $w \in \text{nbr}(u)$; (3) Connectivity to base station is validated in $O(\alpha(V))$ via Disjoint-Set Union with path compression and rank optimization: $\text{find}(w) == \text{find}(v_{\text{BS}})$; (4) Candidate bridge node b minimizing link cost $E_{\text{tx}}(k, d(u, b))$ is spliced into the active path without resetting global tables.

Splicing candidate verification requires $O(\text{deg}(u)\alpha(V))$ time, achieving a $3.9\times - 6.1\times$ execution speedup over full DP recomputation ($884.73\ \mu\text{s}$ vs $3443.04\ \mu\text{s}$) while maintaining 100% packet delivery.

C. Probabilistic Belief-State Routing

To prevent excessive telemetry radio overhead, motes report battery levels every R rounds ($R \in [1, 25]$). Between reporting epochs, the router tracks battery uncertainty via a 24-bin discrete Histogram Belief Filter: (1) **Boundary Clipping:** Exact probability mass accumulation at 0 (brownout) and C_i (saturation); (2) **Sheppard's Correction:** Adjusts variance by $\delta_{\text{bin}}^2 / 12$; (3) **Log-Domain Max-Product DP:** Maximizes log-confidence over path hops and waiting delays $w \in [0, D]$; (4) **Sequential Load Reservations:** Deducts virtual reserved energy from intermediate relays.

IV. EXPERIMENTAL VERIFICATION & RESULTS

A. Automated Pytest Verification Suite (75/75 Passing)

TABLE II: Automated Pytest Verification Suite Breakdown (75 Tests Passing in 1.51s)

Test Module	Target Algorithms & Invariants Verified	Tests	Status
`test_energy_model.py`	1st-order dissipation: free space d^2 cost, multipath d^4 cost, crossover d0 ≈ 87.7m.	4	**PASSED**
`test_adversarial_counterexample.py`	5-node graph isolating lookahead: Time-DP chooses recharging path; Dijkstra fails.	1	**PASSED**
`test_dp_lifetime.py`	3D table filling, boundary clamping at E_max, zero-harvest fallback equivalence.	5	**PASSED**
`test_rip_up_reroute.py`	Union-Find path compression, union by rank, local bridge detour splicing.	4	**PASSED**
`test_dsu_speedup.py`	DSU detour latency reduction (5.5x) and 0% packet loss verification.	1	**PASSED**
`test_uncertainty_routing_and_calibration.py`	24-bin histogram calibration, Sheppard's variance, Brier score, ECE, DP recharge waiting.	7	**PASSED**
`test_battery_heterogeneity_and_fading.py`	Initial battery heterogeneity, log-normal shadow fading (sigma=0..5dB), Sigmoid PRR, MAC retries, and ADC noise.	7	**PASSED**
`test_energy_view.py`	EnergyView protocol conformance across Oracle, Blind, Stale, Gaussian views.	5	**PASSED**
`test_clustering.py`	LEACH and EH-LEACH election probability weighting and non-empty CH sets.	4	**PASSED**
`test_harvesting_model.py`	Solar diurnal day/night cycles, Poisson arrivals, canopy shadow attenuation.	6	**PASSED**
`test_backend_api.py`	FastAPI routes (/health, /simulate, /benchmark, /runs, CSV exports, bounds rejection).	10	**PASSED**
`test_network.py, Baselines`	Adjacency list construction, Dijkstra, A* search, real-trace solar replay.	21	**PASSED**
Total Verification Suite	**Entire Algorithmic and Full-Stack Core**	**75**	**100% PASS**

B. Multi-Seed Statistical Benchmark (N = 30 Seeds, 350 Rounds)

TABLE III: Multi-Seed Statistical Benchmark Results Across Regimes (N = 30 Seeds, 350 Rounds)

Simulation Regime & Configuration	First Node Death (FND)	Alive Nodes (Round 350)	Total Residual Energy (Mean ± Std [95% CI])
1. Baseline (No Harvesting)	89.2 ± 5.4	0.4 ± 0.5	0.0001 ± 0.0002 J [0.0000, 0.0002]
2. Solar (Unaware LEACH + Dijkstra)	351.0 ± 0.0	50.0 ± 0.0	**1.7050 ± 0.0813 J** [1.6759, 1.7341]
3. Solar (Adaptive Time-DP)	351.0 ± 0.0	50.0 ± 0.0	1.6299 ± 0.0939 J [1.5963, 1.6635]
4. Shadowed Solar (Unaware Dijkstra)	**110.2 ± 9.6**	**24.1 ± 3.2**	**0.3900 ± 0.1996 J** [0.3186, 0.4614]
5. Shadowed Solar (Time-DP)	109.0 ± 9.5	23.7 ± 3.8	0.3521 ± 0.1958 J [0.2820, 0.4222]
6. Stochastic Poisson (Unaware LEACH)	351.0 ± 0.0	50.0 ± 0.0	**15.8599 ± 0.1470 J** [15.8073, 15.9125]
7. Stochastic Poisson (Adaptive Time-DP)	351.0 ± 0.0	50.0 ± 0.0	15.7890 ± 0.1527 J [15.7343, 15.8437]

C. Energy Uncertainty Routing Benchmark Across Solar Regimes

TABLE IV: Energy Uncertainty Routing Benchmark Across Solar Regimes (180 Runs)

Regime	Routing Method	PDR	Hops	Wasted E	Brier	ECE	Oracle Gap Closed
Poor	M0 (Blind Dijkstra)	47.50%	2.62	0.0268 J	0.0262	0.0262	Baseline
Poor	M1 (Stale Dijkstra, R=20)	46.03%	2.40	0.0212 J	0.0394	0.0394	-17.3%
Poor	**M3 (Belief DP, Proposed)**	**47.47%**	2.47	0.0227 J	0.0272	0.0230	**+1.44% vs M1**
Poor	**M4 (Belief LCB-Maximin)**	**52.19%**	3.04	0.0216 J	0.0233	0.0192	**+55.4% Closed**
Poor	M2 (Oracle Upper Bound)	55.96%	3.46	0.0145 J	0.0000	0.0000	100.0%
Balanced	M0 (Blind Dijkstra)	60.73%	2.66	0.0088 J	0.0049	0.0049	Baseline
Balanced	M1 (Stale Dijkstra, R=10)	67.81%	2.69	0.0009 J	0.0046	0.0046	+120.1%
Balanced	**M3 (Belief DP, Proposed)**	**65.53%**	2.70	0.0027 J	0.0019	0.0022	**+81.5% Closed**
Balanced	M2 (Oracle Upper Bound)	66.63%	3.33	0.0000 J	0.0000	0.0000	100.0%

D. DSU Detour Speedup, Scalability & Habitat Deployment

• **DSU Detour Speedup:** Across 500 fault injection trials, DSU local detour repair sustained a 100.0% packet delivery ratio up to a 30% node failure rate, executing in 884.73 µs compared to 3443.04 µs for full DP recomputation (a **3.9× - 6.1× latency reduction**).
• **Embedded Scalability:** Across network scales N = 50 to 500 nodes, 3D Time-DP computes in 2.25 ms to 7.76 ms (sub-8 ms), confirming real-time execution feasibility on 32-bit ARM Cortex-M MCUs.
• **Great Duck Island Habitat Case Study:** In an empirical 32-mote deployment under 60% spruce canopy shade over 500 rounds, LEACH suffers First Node Death at round 433 and EH-LEACH at round 445. Our proposed Time-DP + DSU sustains **100% mote survival with zero node deaths (FND ≥ 500)**, achieving a **+15.5% mission lifetime extension**.

V. ENGINEERING LIMITATIONS & THREATS TO VALIDITY

1) **Single-hop vs Multi-hop Crossover:** Electronic reception overhead ($E_{\text{rx}} = 0.20\text{ mJ}$) causes multi-hop forwarding to consume more energy than direct single-hop transmission when distances are short ($d < 87.7\text{ m}$). Multi-hop lookahead is justified when distance exponents (d^4) or severe obstacles dictate.
2) **Timescale Decoupling:** Transmission delays occur at millisecond resolution while solar variation occurs at minute/hour scales, synchronized by discrete slot offsets.
3) **Hardware Testbed Deployment:** While verified through 75 automated unit tests and a full-stack software implementation, physical validation on TelosB / CC2650 motes under outdoor sunlight remains future laboratory work.

VI. CONCLUSION

This paper presented an end-to-end, deterministic algorithmic framework for Energy-Harvesting Wireless Sensor Networks. By combining 3D Time-Augmented Dynamic Programming, Disjoint-Set Union (DSU) live detour recovery, and a 24-bin discrete Histogram Bayesian Filter, the proposed system eliminates the false rejection of recharging relays, achieves live fault tolerance in $O(\text{deg}(u)\alpha(V))$ time, and closes up to 81.5% of the oracle delivery gap under stale telemetry. Future research will focus on physical Contiki-NG deployments on hardware motes in the SELECT laboratory.

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